based on antibodies are not currently able to identify exclusively the free (active) regulatory subunits, as can the biologically active direct cytochemical probe, described here. Chiefly this is because all currently available antibodies recognize C and R subunits in the free as well as holoenzyme form. Thus, they cannot be used to monitor net changes in the amount or location of free subunits upon stimulation of cells by agents that act via cAMP. This constraint for immunologic approaches may be unique to studies of the cAMP-dependent protein kinases since no other known regulatory enzyme requires dissociation of its subunits to become active. Antibodies to the protein kinases can, however, discriminate type I from type II R subunits, which the F:C probe cannot do.

When prepared and used as described in this and the previous chapter the molecular probes for localizing R subunit and C subunit[12,13] can yield data demonstrating protein kinase activities in individual cells. In conjunction with immunocytochemistry and conventional biochemistry this approach should be useful in studying cell regulatory events mediated by cAMP.

[25] Identification and Ligand-Induced Modification of the cAMP Receptor in *Dictyostelium*

By PETER KLEIN, ANNE THEIBERT, and PETER DEVREOTES

I. Introduction

Early in the development of *Dictyostelium discoideum,* an intercellular communication system appears which coordinates the highly organized aggregation of one million cells. Two responses essential to this coordinated aggregation are chemotaxis to cAMP and cAMP signaling (the synthesis and secretion of cAMP in response to extracellular cAMP), both of which are mediated by a cell surface cAMP receptor. These response systems provide excellent models for transmembrane signaling in eukaryotic cells.

Most of the known components of these response systems are under tight developmental regulation. For example, receptor binding activity is low in growing cells, increases to a maximum of about 10^5 sites/cell at 5 to 6 hr of development as the cells are spontaneously aggregating, and declines to a low level by 9 to 10 hr as multicellular structures assemble (Fig. 1).

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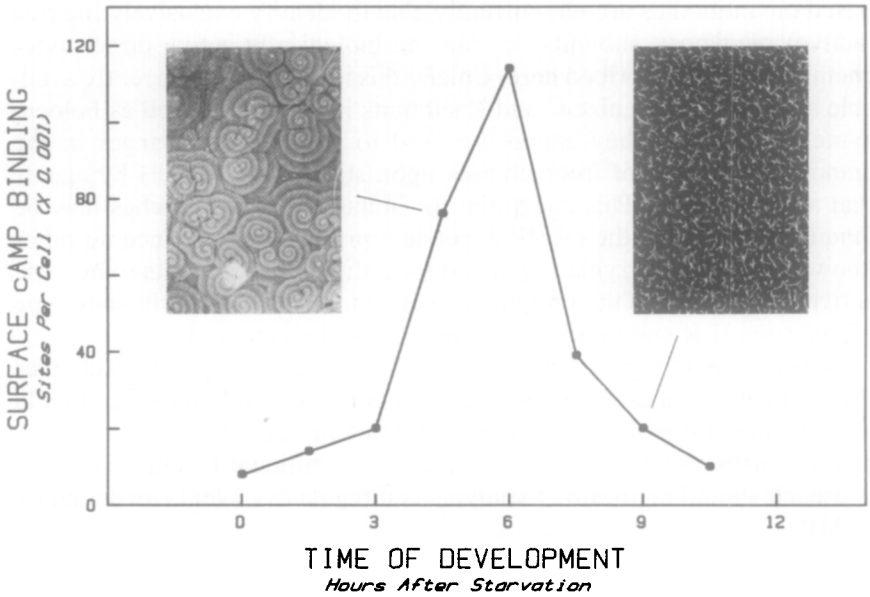


FIG. 1. Expression of surface cAMP-binding sites during development: Cells were freed of nutrients (at $t = 0$) and starved in suspension as described in Methods. At the times indicated an aliquot was removed and the level of [^{3}H]cAMP binding was determined. Photographs show the appearance of 4- and 7-hr cells plated on a monolayer. Territories in the 4-hr samples are 1–2 cm in diameter. Aggregates in 7-hr samples are 1–2 mm.

Most of the cellular responses to cAMP subside within minutes when the stimulus concentration is held at a fixed level. This reversible adaptation to cAMP, which has been extensively studied for cAMP signaling (for a review, see Ref. 1), could involve covalent modification of the cAMP receptor, which may be similar to mechanisms of adaptation postulated for other signal transduction systems such as methylation of bacterial chemotactic receptors and phosphorylation of rhodopsin in rod outer segments.

Using the photoreactive cAMP analog 8-N_3-[^{32}P]cAMP, the cAMP receptor has been identified as a doublet of M_r 40,000–43,000.[2] The doublet represents interconvertible bands that shift electrophoretic mobility in response to cAMP.[3] The shift in mobility, which is likely caused by covalent modification of the receptor, shows kinetics and cAMP dose

[1] P. Devreotes, *in* "The Development of *Dictyostelium*" (W. F. Loomis, ed.), p. 117. Academic Press, New York, 1982.

[2] A. Theibert, P. Klein, and P. Devreotes, *J. Biol. Chem.* **259,** 12318 (1984).

[3] P. Klein, A. Theibert, D. Fontana, and P. Devreotes, *J. Biol. Chem.* **260,** 1757 (1985).

dependence that closely paralleled those observed for adaptation of the physiological responses.[4] Further, the shift to lower mobility is correlated with a 6-fold increase in phosphorylation of the receptor.[5]

This chapter focuses on methods to identify the cAMP receptor and to monitor the cAMP-induced modification by specific photoaffinity labeling and by *in vivo* phosphorylation.

II. Photoaffinity Labeling with 8-N_3-[^{32}P]cAMP

Photoaffinity labeling of nucleotide binding proteins in general and, specifically, cAMP-binding proteins (i.e., the regulatory subunit of cAMP-dependent protein kinases) has been described.[6,7] There are also several reports on photoaffinity labeling of cAMP-binding proteins in *Dictyostelium* with 8-N_3-[^{32}P]cAMP.[2,8-10] However, photoaffinity labeling of the surface cAMP receptor[2,9,10] poses special problems: (1) the receptor–cAMP complex has a very rapid off rate which precludes washing the cells after ligand binding, thereby lowering the specificity of the photoreaction; (2) there are several cAMP-binding proteins in *Dictyostelium* and, in particular, the regulatory subunit of cAMP-dependent protein kinase has a similar electrophoretic mobility (M_r 41,000) and is readily labeled with 8-N_3-[^{32}P]cAMP[8]; (3) *Dictyostelium* phosphodiesterase (PDE) avidly hydrolyzes 8-N_3-cAMP; (4) cells contain abundant proteolytic activity and the cAMP receptor is extremely sensitive; (5) the receptor has a marked tendency to aggregate irreversibly. Methods to overcome each of these difficulties are discussed.

The synthesis of 8-N_3-cAMP has been described in detail elsewhere.[11] The ^{32}P analog is available from ICN on a monthly basis (specific activity = 60–70 Ci/mmol) and the ^{3}H derivative is available from NEN (10–20 Ci/mmol). As described in Ref. 6, the azido compound is stable up to 4 weeks if stored as the triethylammonium salt in methanol at −20° in the dark (as supplied by ICN). The compound is sensitive to higher temperatures and should be kept at −20° except for removal of sample. It is also reportedly

[4] P. Devreotes and J. Sherring, *J. Biol. Chem.* **260,** 6378 (1985).
[5] P. Klein, D. Fontana, A. Theibert, B. Knox, and P. Devreotes, *Cold Spring Harbor Symp. Quant. Biol.* **50,** 787–799 (1985).
[6] R. L. Potter and B. E. Haley, this series, Vol. 91, p. 613.
[7] U. Walter and P. Greengard, this series, Vol. 99, p. 154.
[8] B. H. Leichtling, I. H. Majerfeld, E. Spitz, K. L. Schaller, C. Woffendin, S. Kakinuma, and H. V. Rickenberg, *J. Biol. Chem.* **259,** 662 (1984).
[9] M. Juliani and C. Klein, *J. Biol. Chem.* **256,** 613 (1981).
[10] B. L. Meyers-Hutchins and W. A. Frazier, *J. Biol. Chem.* **259,** 4379 (1984).
[11] B. E. Haley, this series, Vol. 46, p. 339.

sensitive to acid.[6] The compound is stable to brief exposure to fluorescent room lights. To verify purity and photoreactivity of each new shipment, the compound should be analyzed by PEI thin-layer chromatography run with 2 *N* formic acid, pH 3.4, prior to and following extensive irradiation. If necessary, the azido cAMP can be further purified by ion-exchange chromatography on Dowex-50.

III. Methods

A. Preparation of Cells

Development. Ax-3 cells of *Dictyostelium discoideum* are grown in 400 ml of HL-5 medium[12] to a density of 5×10^6/ml. To initiate development, cells are centrifuged (600 *g*, 5 min), resuspended in an equal volume of development buffer (DB) (5 m*M* Na_2HPO_4, 5 m*M* NaH_2PO_4, 2 m*M* $MgSO_4$, 200 μM $CaCl_2$), centrifuged again, resuspended at 2×10^7/ml in DB (100 ml), and shaken at 110–130 rpm in a 1-liter Erlenmeyer flask. Spontaneous oscillations in cellular cAMP synthesis begin at 3.5 hr and continue through 7 hr. Early developmental markers, such as adenylate cyclase and surface cAMP receptors, reach peak expression (at least 12-fold above growth levels) within 5 to 6 hr (Fig. 1).

B. Photoaffinity Labeling

Cells are prepared at the appropriate time(s) in development, washed once in DB, once in DB with 10 m*M* DTT, and then resuspended in DB/DTT at 10^8/ml at 0° (DTT is included to inhibit PDE). The 8-N_3-[^{32}P]cAMP (60–70 Ci/mmol) is dried from methanol under an N_2 stream and resuspended in DB/DTT at 0.75 μM. In a 12 × 75 mm glass test tube, 100 μl of cells is added to 100 μl of 8-N_3-[^{32}P]cAMP. The N_3-cAMP solution should be used immediately since DTT reduces the azido group.[5] In a parallel tube, nonradioactive cAMP is added at 100 μM to determine nonspecific binding. The reaction is vortexed and after 30 sec, 3 ml of 98% saturated ammonium sulfate (AS) is added.[13] In the presence of AS, the half-time for dissociation of the receptor–cAMP complex is increased from a few seconds to several hours. After 5 min (to allow complete stabilization of binding), cells are centrifuged at 4000 rpm for 10 min (Sorvall HS-4). The supernatant is discarded and the cells are washed again in AS. The cell pellet after the second AS wash is resuspended at a density of 10^8/ml in

[12] D. J. Watts and J. Ashworth, *Biochem. J.* **119,** 171 (1970).
[13] P. Van Haastert and E. Kien, *J. Biol. Chem.* **258,** 9636 (1983).

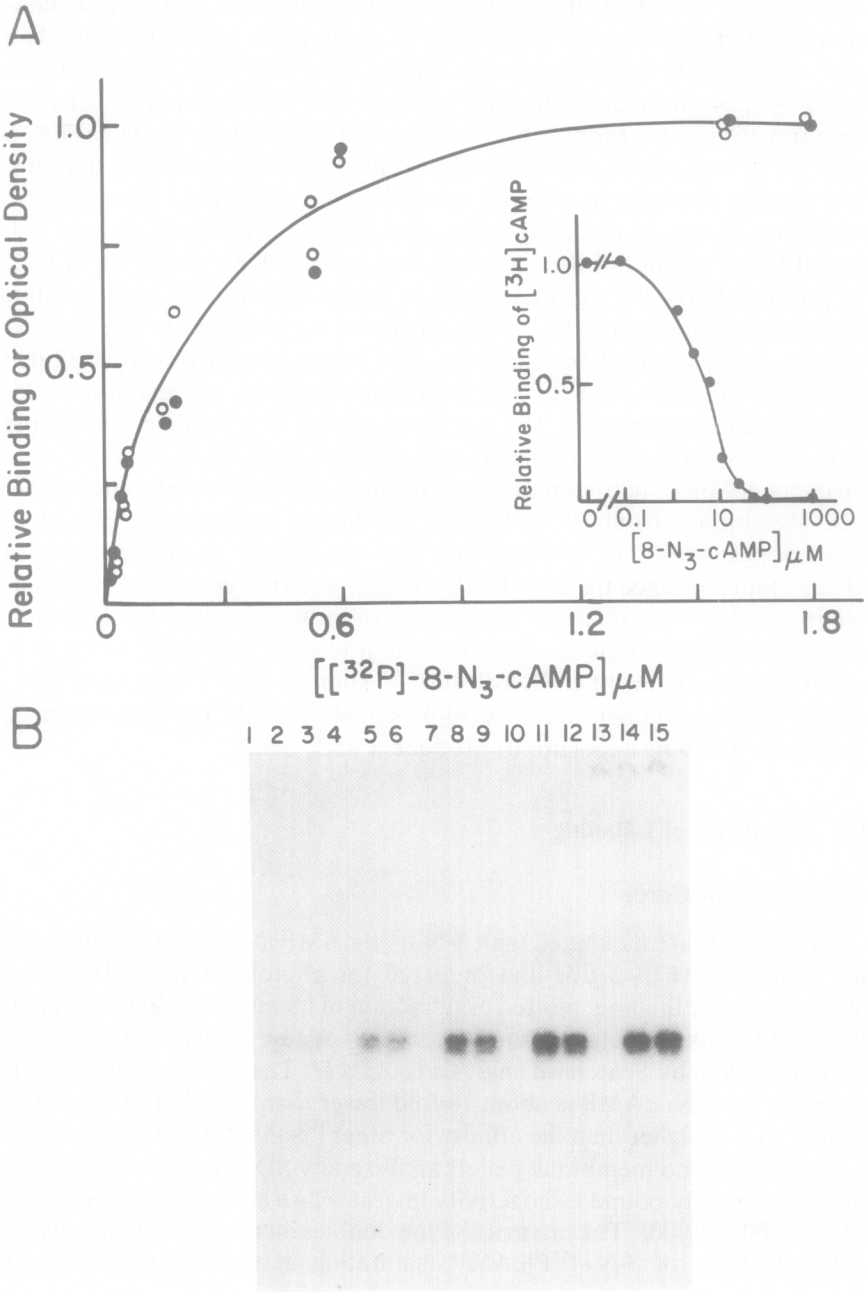
AS (200 μl). A small aliquot (10–20 μl) is removed to measure noncovalent binding and the remainder is spotted on a plastic Petri dish at 4°. The sample is irradiated with a short wave UV lamp (Mineral Light UVG-54, 254 nm) from a distance of 5 cm for 5 to 20 min. The cells are then transferred to receptor buffer (25 m*M* Tris, pH 7.5, 5 m*M* EDTA, 5 m*M* DTT, with protease inhibitors described below) at a 10-fold dilution. The sudden decrease in osmotic strength causes cell lysis. The protease inhibitors (1 mg/ml phenanthroline, 250 μg/ml CBZ-phenylalanine, 100 μg/ml PMSF, 1 mg/ml benzamidine, 10 μg/ml TPCK, 10 μg/ml TLCK, 10 μg/ml TAME, 0.5 μg/ml antipain, 1 μg/ml leupeptin, 0.5/ml aprotin, 0.6 μg/ml chymostatin, 0.6 μg/ml pepstatin) are required to prevent extensive degradation of the receptor. The lysate is centrifuged at 12,000 rpm for 30 min (Sorvall SS-34); after centrifugation, no receptor is detected in the supernatant. The membrane pellet is solubilized in sample buffer (1.5% SDS, 62.5 m*M* Tris, pH 6.8, 5% glycerol, 2.5% DTT, and bromphenol blue) in a volume equal to the original reaction volume (200 μl). Membranes from 10^7 cells must be diluted into at least 200 μl to prevent aggregation of the receptor. The samples are not boiled prior to electrophoresis since this also tends to aggregate the receptor. Once solubilized in SDS, the receptor is stable at room temperature for at least 3 days and can be stored at −20° for several weeks. The samples are then electrophoresed on 10% SDS–PAGE,[14] stained with Coomassie blue, dried, and autoradiographed. A 50-μl sample usually requires no more than a 15- to 20-hr exposure to produce a dark band.

IV. Specificity of Labeling

A. Saturation Curve

Intact cells are incubated with 8-N_3-[^{32}P]cAMP at concentrations ranging from 10 n*M* to 2 μ*M* and prepared for photolabeling as described above. Specific binding, plotted as a fraction of maximal binding, is shown in Fig. 2A (closed circles). Saturation occurs at approximately 1 μ*M*. The K_D determined by Scatchard analysis is 0.3 μ*M*. The affinity of the cAMP receptor for 8-N_3-cAMP is about 10-fold lower than for cAMP (K_D = 30–50 n*M*) but is higher than the affinity for other 8-substituted analogs. Cells are irradiated and membrane pellets analyzed by SDS–PAGE. About 90% of the covalently bound radioactivity migrates as a closely spaced doublet (M_r 40,000–43,000). The intensity of the doublet increases with increasing concentrations of 8-N_3-[^{32}P]cAMP, saturating at approximately 1 μ*M*

[14] U. Laemmli, *Nature (London)* **227,** 680 (1970).


A
Relative Binding or Optical Density
1.0
0.5
0
0.6
1.2
1.8
[[32P]-8-N3-cAMP] μM
Relative Binding of [3H]cAMP
1.0
0.5
0
0.1
10
1000
[8-N3-cAMP] μM
B
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

(Fig. 2B, lanes 2, 3, 5, 6, 8, 9, 11, 12, 14, 15). Addition of 100 μM cAMP to the binding assay completely inhibits photolabeling of the doublet (Fig. 2B, lanes 1, 4, 7, 10, 13). Quantification of scans of autoradiographs in Fig. 2B are plotted in Fig. 2A (open circles). The saturation curves for noncovalent binding and photoaffinity labeling of the doublet are identical.

Nonradioactive 8-N_3-cAMP competes the binding of [^{3}H]cAMP (inset, Fig. 2A) in the same concentration range that 8-N_3-[^{32}P]cAMP binds (K_I of about 1 μM). These K_I values are within the reported concentration ranges over which cAMP (1 nM to 1 μM) and 8-N_3-cAMP (approximately 1 to 10 μM) stimulate chemotaxis and cAMP signaling.[15] The total number of sites (about 10^5/cell) was the same when determined by [^{3}H]cAMP or 8-N_3-[^{32}P]cAMP binding. In all of these determinations, the optical densities of the higher and lower mobility bands of the doublet were considered together. However, there is little apparent difference in the relative intensities of the two bands of the doublet at any level of saturation or competition. It was subsequently determined that the K_D values for binding of [^{3}H]cAMP to the higher and lower mobility forms are 33 and 13 nM, respectively.[4]

B. Inhibition by cAMP

Noncovalent binding of 8-N_3-[^{32}P]cAMP is competed by cAMP in the 10 nM to 1 μM range (Fig. 3A, closed circles). cAMP inhibits photolabeling in the same concentration range; quantification of scans of the autoradiographs are plotted in Fig. 3A (open circles). The cAMP competition curves for noncovalent binding and photoaffinity labeling are identical. The K_I for cAMP inhibition of both photolabeling and noncovalent binding was about 70 nM (Fig. 3A), which is similar to the K_D reported for [^{3}H]cAMP binding (30–50 nM).

[15] L. J. Wallace and W. A. Frazier, *Proc. Natl. Acad. Sci. U.S.A.* **76,** 4250 (1979).

FIG. 2. Saturation curve: (A) Cells were incubated with 8-N_3-[^{32}P]cAMP at 18.5 nM to 1.8 μM. Binding was determined as described above and was normalized to maximal binding (closed circles). Maximal binding was typically about 10^5 sites/cell (80,000 cpm/10^7 cells). (B) Cells were then irradiated and membranes were isolated and run on 10% SDS-PAGE. A representative autoradiogram is shown with ligand present at 20 nM (lanes 1–3), 55.6 nM (lanes 4–6), 0.167 μM (lanes 7–9), 0.5 μM (lanes 10–12) and 1.5 μM (lanes 13–15). The major band migrates as a doublet with an apparent molecular weight of 40,000 to 43,000. Lanes 1, 4, 7, 10, and 13 show photolabeling when 100 μM cAMP was included in the binding assay. Autoradiograms were scanned and the optical density in the doublet (M_r 40,000–43,000) was normalized as a fraction of maximal optical density [open circles in (A)]. Inset shows the competition of [^{3}H]cAMP binding by 8-N_3-cAMP.

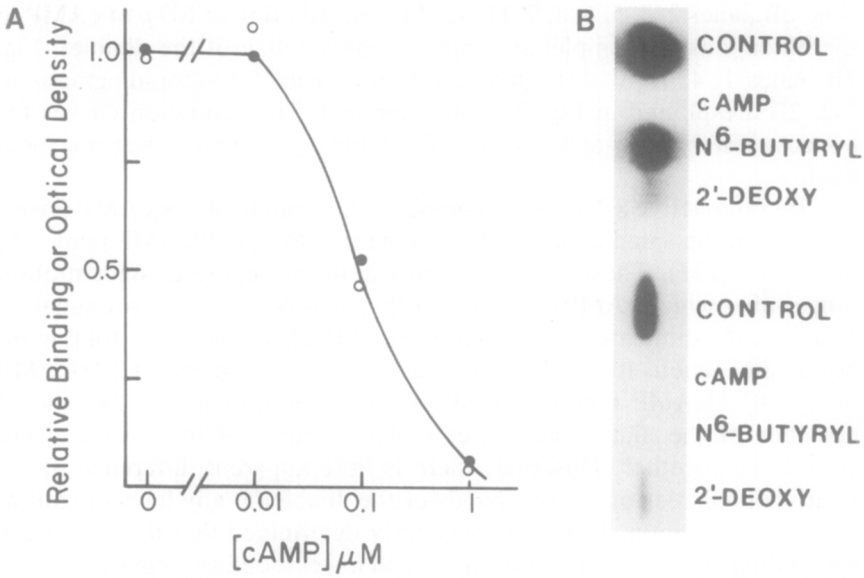


FIG. 3. Inhibition by cAMP and analogs: (A) Cells were incubated with 8-N_3-[^{32}P]cAMP at 0.2 to 0.3 μM final concentrations with nonradioactive cAMP present at 10 nM, 0.1 μM, and 1 μM and binding was determined and plotted as described above (closed circles). Cells were irradiated and membranes were analyzed on SDS-PAGE as above. The autoradiographs were scanned and plotted (open circles) as the fraction of binding in the absence of cAMP. (B) Competition of photoaffinity labeling of the receptor (left) and the regulatory subunit of cAMP-dependent protein kinase (right). Photolabeling of the cAMP receptor was carried out as described in Methods. Photolabeling of the regulatory subunit of protein kinase was carried out as described in Ref. 7. cAMP, N^6-monobutyryl-cAMP, and 2′-H-cAMP were included in the labeling reactions at concentrations of 100, 40, and 1.0 μM, respectively.

C. Analog Specificity

The several cAMP-binding proteins in *Dictyostelium* can be distinguished pharmacologically. For example, 2′-deoxy-cAMP has a high affinity for the surface receptor but not for the regulatory subunit of cAMP-dependent protein kinase whereas N^6-monobutyryl-cAMP has a high affinity for the regulatory subunit but does not bind to the surface receptor even at 0.1 mM. Figure 3B illustrates competition with these two analogs of the photolabeling of the receptor and the regulatory subunit of the cAMP-dependent protein kinase. These differences in analog specificity have been used recently in studies of cAMP-mediated induction of gene expression during development. It appears that the surface receptor,

rather than the regulatory subunit of the kinase, is the immediate target for cAMP effects on gene expression.[16]

V. Receptor Modification

A. Electrophoretic Mobility Shift

A reversible shift in electrophoretic mobility of the receptor has been shown to be stimulated by binding of extracellular cAMP.[3,4] The higher mobility from the receptor, found prior to cAMP stimulation, is designated R (M_r 40,000) and a lower mobility form, found after cAMP stimulation, is designated D (M_r 43,000). In order to preferentially label either the R or D form of the receptor, aggregation stage cells are washed once in DB, resuspended at 10^8/ml at 22°, and shaken at 150 rpm. To label the R form, cells are treated with 5 m*M* caffeine for 30 min, which blocks activation of adenylate cyclase.[17] To label the D form, cells are treated with 10 m*M* DTT (to inhibit PDE) followed by stimulation with 10 μM cAMP for 13 min. Concanavalin A (100 μg/ml) is then added (since it improves recovery of receptor) for 2 min, then cells are washed in DB and then in DB/DTT at 0°, resuspended at 2×10^8/ml in DB/DTT, and photolabeled as described above. Frequently, the R form of the receptor appears more intense than the D form from a parallel set of cells, even when the level of noncovalent binding of the photolabel is similar. This is apparently due to selective degradation of the D form of the receptor during membrane preparation and is also observed in Western blots.

The concentration dependence and kinetics of receptor modification correlate closely with those of adaptation of several physiological responses, including cAMP signaling, myosin phosphorylation, and cell shape change. The concentration dependence of the steady state distribution of the R and D forms of the receptor is shown in Fig. 4A (closed circles). In cells pretreated with caffeine (so that the external cAMP concentration is low), only 10% of receptors are found in the D form (Fig. 4A, lanes 1 and 2). A detectable increase in the fraction of receptors in the D form is observed with 0.5 n*M* cAMP; 50% of the maximal shift occurs with 30 n*M* cAMP; and the maximal fraction in the D form, about 80%, occurs with 1 μM cAMP. The concentration of cAMP that induces a half-maximal shift in the distribution of receptor forms is roughly equal to the K_D for cAMP binding. When the cAMP concentration is subsequently

[16] P. Schaap and L. Van Driel, *Exp. Cell Res.* **159,** 388 (1985).
[17] M. Brenner and S. Thoms, *Dev. Biol.* **101,** 136 (1984).

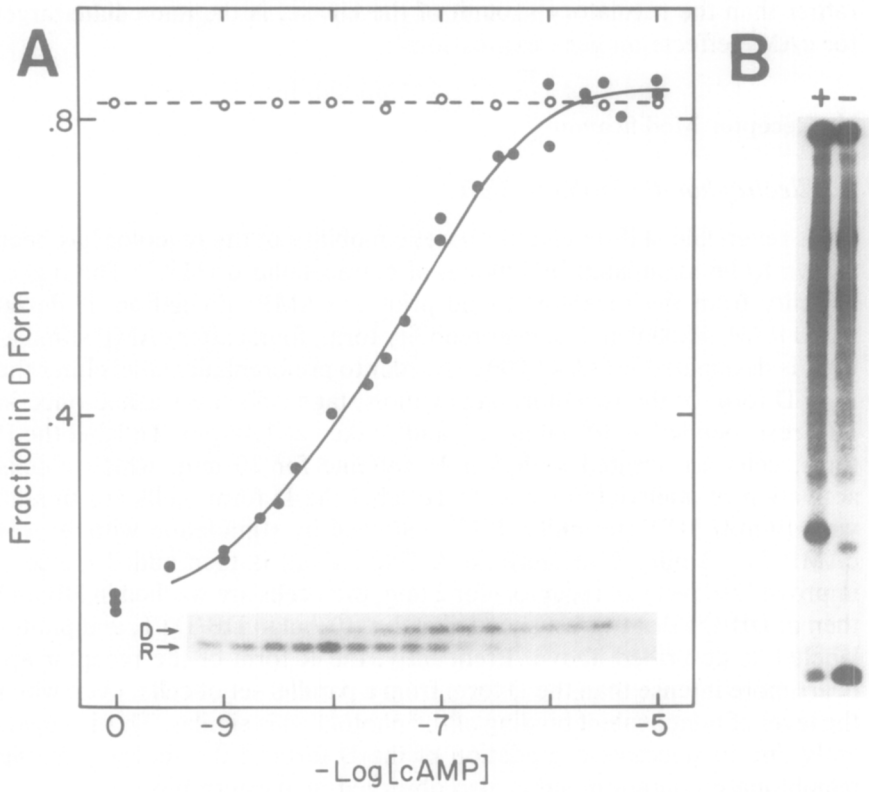


FIG. 4. (A) Dose dependence of receptor modification: Cells were stimulated with the indicated concentrations of cAMP for 15 min (closed circles), washed, and photolabeled as described above. The inset shows a portion of the gel (10% total acrylamide with 0.05% bisacrylamide). The first 2 lanes are controls and the remaining 14 lanes (from right to left) represent a 2-fold dilution series starting at 10 μM cAMP. After aliquots were taken at 15 min, 10 μM cAMP was added to each set of cells. After an additional 15 min aliquots were taken for analysis (open circles). (B) *In vivo* phosphorylation of the receptor: Cells were labeled with [^{32}P]phosphate for 30 min, divided, and incubated in the absence (−) or presence (+) of 10 μM cAMP for 15 min. Membranes were prepared and separated by SDS/PAGE (10% total acrylamide with 0.05% bisacrylamide). The R and D forms (M_r 40,000–43,000) are apparent in the − and + lanes, respectively.

increased from any concentration to 10 μM and the incubation continued for 15 min, the fraction of receptors in the D form rises to about 80% in all cases (Fig. 4A, open circles).

The cAMP-triggered redistribution of receptors in each form occurs with a half-time of about 2 min. By 15 min, a plateau value (80% in the D form) is reached which remains constant during persistent stimulation; 50 nM cAMP, a concentration which occupies about 50% of the surface

receptors, also induces a rapid redistribution which occurs with a half-time of about 2 min but the final plateau value reached is only 40% in the D form. When cells are washed free of cAMP and incubated at 22°, there is a return toward the basal state which occurs with a half-time of 5–6 min. This reversal is completely blocked at 0°.[4]

B. In Vivo Phosphorylation

The shift to the D form is associated with a 6-fold increase in the level of phosphorylation of the receptor.[5] To phosphorylate the receptor *in vivo,* cells at the aggregation stage are washed twice in 10 m*M* MES, pH 6.8, resuspended at 10^8/ml in 10 m*M* MES, and shaken at 150 rpm. Carrier-free $^{32}PO_4$ (1.0 mCi/ml of cells at 10^8/ml) is added for 30 min. For the R form, 5 m*M* caffeine is added at the same time. For the D form, DTT and cAMP are added 15 min after addition of $^{32}PO_4$. Con A is added and membranes are prepared as described above. When endogenous cAMP synthesis is suppressed, the R form of the receptor predominates in autoradiographs of phosphorylated or photolabeled cells (Fig. 4B, lane −). After stimulation with cAMP, the D form predominates (Fig. 4B, lane +).

C. Resolving R and D Forms by SDS/PAGE

The resolution of the two forms of the receptor depends markedly on the concentration of bisacrylamide in the gel. Furthermore, the mobility of both forms of the receptor relative to other proteins (e.g., actin) also varies with the concentration of bisacrylamide. For example, using a 10% gel with 0.05% bis, the two forms of the receptor are widely separated (see Fig. 4A and B). The D form of the receptor migrates slower than actin while the R form migrates faster. Using a 10% gel with 0.25% bis the two forms migrate more closely together (see Fig. 2B and Fig. 3B) and both have a slightly higher mobility than actin. On a 10% gel with 1% bis, the two forms have very similar mobilities but a significantly higher mobility than actin. Due to this anomalous behavior, a gel with 1% bis resolves the receptor from contaminants that coelectrophorese on the 0.05% bis gels, and, in fact, two-dimensional preparative gels using two different bis concentrations are useful in purification of the receptor (manuscript in preparation).

VI. Summary

Photoaffinity labeling with 8-N_3-[^{32}P]cAMP and *in vivo* phosphorylation with $^{32}PO_4$ provide sensitive methods for the identification of the surface cAMP receptor. Photoaffinity labeling achieves high specificity

while, in practice, the *in vivo* ^{32}P labeling affords considerably higher sensitivity. Both methods allow fairly direct assays of cellular responses to an extracellular cAMP stimulus at the level of the receptor—a shift in electrophoretic mobility and a rise in phosphorylation. Quantification of the shift in mobility, assayed by photolabeling, has shown a strong correlation between modification of the receptor and adaptation.[4] These labeling techniques will also help to elucidate the role of the concomitant changes in receptor phosphorylation, which may cause the change in electrophoretic mobility. Furthermore, photolabeling and ^{32}P labeling will allow study of the possible multiple affinity classes of the cAMP receptor[18] and of the possible expression of cAMP receptors in later stages of development.[19]

Acknowledgments

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[18] P. J. M. Van Haastert and R. J. W. De Wit, *J. Biol. Chem.* **259,** 13321 (1984).
[19] P. Schaap and W. Spek, *Differentiation* **27,** 83 (1984).

[26] Crystallizing Catabolite Gene Activator Protein with cAMP for Structural Analysis

By IRENE T. WEBER

The tertiary structure of a protein is essential for its biological function. X-Ray crystallography is the only method currently available to determine the three-dimensional positions of the atoms in a protein molecule.[1] The crystal structure provides valuable information on the interaction between the protein and ligands. Catabolite gene activator protein (CAP) from *Escherichia coli* has been crystallized in a complex with cAMP[2] and the structures of the wild type[3,4,19] and of a mutant, CAP91,[20] have been determined.

[1] T. L. Blundell and L. N. Johnson, "Protein Crystallography." Academic Press, New York, 1976.
[2] D. B. McKay and M. G. Fried, *J. Mol. Biol.* **139,** 95 (1980).
[3] D. B. McKay and T. A. Steitz, *Nature (London)* **290,** 744 (1981).
[4] D. B. McKay, I. T. Weber, and T. A. Steitz, *J. Biol. Chem.* **257,** 9518 (1982).

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